Md. Irtiza Hossain

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Personal Website md-irtiza-hossain irtiza

Passionate about advancing research in AI and its applications in security, I bring strong expertise in machine learning, image processing, computer vision, and system security. With a solid foundation in Python, TensorFlow, and full-stack development, I combine research experience with practical software engineering skills. I am currently engaged in an academic role and eager to contribute to innovative AI and security projects in a PhD program.

Professional Experience

Adjunct Lecturer — *Dhaka, Bangladesh*
Brac University — *Nov 2024 – Present*

- Delivered lectures and conducted labs on Data Structures, Artificial Intelligence, Computer Interfacing, Database Management Systems, System Analysis and Design, and Introduction to Robotics.

Junior Software Engineer — *Dhaka, Bangladesh*
Orbit Informatics — *Aug 2024 - Nov 2024*

- Developed a Chrome extension using React for the frontend and Node.js for the backend, enabling seamless browser integration and dynamic user interactions.

Trainee Engineer .NET — *Dhaka, Bangladesh*
PRAN-RFL Group (othoba.com) — *Jun 2024 - Aug 2024*

- Automated the abandoned cart detection system to identify customers who left items in their cart, leading to targeted recovery efforts and a 1–2% increase in sales.

Undergraduate Teaching Assistant — *Dhaka, Bangladesh*
BRAC University — *Oct 2022 - May 2024*

- Assisted in Programming Language-II (Object-Oriented Programming) course (Feb 2023 – May 2024) and Differential Calculus and Coordinate Geometry (MAT 110) course (Oct 2022 - Dec 2022)

Research Works

- **Enhanced CNN Approaches for Multi-Image Embedding in Image Steganography [ICICT 2024]** — Oct 2024

  Md. Irtiza Hossain, Samiul Kadir, Farhan Ishraq Fagun, Ishtiaq Samiul, Rafi Zaman Saukhin — 10.1109/ICICT64387.2024.10839645 — Code

- **Edge-Native Digitization of Handwritten Marksheets: A Hybrid Heuristic-Deep Learning Framework [Qpain 2026]** — Apr 2026

  Md. Irtiza Hossain, Junaid Ahmed Sifat, Abir Chowdhury, Md. Imran Bin Azad — Accepted in *2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking* — arXiv:2508.16295 — Code

- **GLE-SSD-Net: Uncertainty-Aware Distillation for Efficient LiDAR-Camera 3D Object Detection** — Under Review

- **Uncertainty-Aware and Efficient 3D Object Detection for Autonomous Driving: A Comprehensive Survey** — Preparing to Submit

- **SmokeGraph-Bench and RelSmoke: A Benchmark and Relational Approach for Smoking Behaviour Detection under Extreme Class Imbalance** — Preparing to Submit

Education

- **Brac University** *[Master of Science in Computer Science and Engineering]* *Jun 2025 – Present*
- **Brac University** *[Bachelor of Science in Computer Science and Engineering]* *Dec 2019 – May 2024*
  - CGPA: 3.90 out of 4.00

Projects

- **Explainable AI for Brain Cancer Classification from MRI Scans** *github.com/repo*
- **Image Classification Using CNN (TensorFlow)** *github.com/repo*
- **Telecom Churn Prediction** *github.com/repo*
- **COVID-19 Tracker (React)** *Live Link*
- **Power X Gym (MERN Stack)** *Live Link*
- **Gadget Store (PHP, MySQL)** *github.com/repo*

Technologies

Languages: C, C#, Python, JavaScript, PHP

Technologies: .TensorFlow, Scikit-learn, React, NodeJS, Redux , MongoDB, Firebase, MySQL, .Net

Awards and Achievements

- **V**ice Chancellor's List (5 times) — Recognition for exceptional semester GPA performance published each semester.
- **D**ean's List (2 times) — Awarded for outstanding academic performance based on semester GPA.

References

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